



Thermal stability of metal-ceramic Kovar® /94% alumina joints

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Abstract

Metal-ceramic joints have gained considerable attention over the past two decades due to their widespread applications. However, the thermal cycling stability of such joints remains poorly understood. This study investigates the thermal stability of vacuum-brazed joints between Kovar® alloy and 94% alumina ceramic, fabricated using an Ag-Cu brazing alloy and a titanium foil interlayer. The primary objective is to identify the patterns governing the effect of secondary heating on the air-tightness of these high-temperature-brazed metal-ceramic joints. The joint microstructure was characterized by scanning electron microscopy and energy-dispersive X-ray spectroscopy. The structure was found to consist of a eutectic comprising solid solutions based on copper and silver, along with intermetallic compounds (TiCu and FeTi) and a mixed oxide ($\text{Ti}_x\text{Cu}_x\text{O}$). The necessity of brazing at 850°C for 10 min to achieve hermetic metal-ceramic joints was established. The thermocycling test results demonstrated that the volume of brazing alloy in the joint affects its heat resistance upon reheating. Numerical modeling

confirmed that an increase in the height of the Ag-Cu brazing alloy column elevates the stress intensity and expands the high-stress zones, particularly at the ceramic/brazing alloy interface.



Keywords

Brazing; Alumina; Stresses; Microstructure; Thermocycling; Modeling

1. Introduction

Metal-ceramic joints find applications in various fields due to their unique properties [1]. Ceramics are particularly attractive for their superior corrosion and wear resistance at high temperatures [2], while metals are characterized by high plasticity, impact toughness, and strength [3].

One of the most common applications for metal-ceramic joints is electron signal transmission through a vacuum barrier. The key requirements for such installations are thermal stability and air-tightness. In their work, Share R [4], described examples of such brazed joints and emphasized the importance of their meticulous design. It has been demonstrated that with an increase in the amount of brazing alloy in the clearance, the residual stress levels in the joints also increase. Residual stress also depends on the type of brazing alloy used. Metal-ceramic joints are also widely used as viewports in high-energy facilities. Different technological approaches to the production of sapphire viewports have been studied in Ref. [5] for small modular reactor applications. However, achieving the durability of such viewports at temperature regimes as high as 450°C has so far been challenging. This is directly caused by residual stress resulting from heating and cooling processes. Currently, active work is underway to develop joining technologies for dissimilar materials that allow for stress relaxation. In medicine, a variety of tasks require air-tight metal-ceramic joints. Modern achievements and approaches to the production of such joints have been described in detail in Ref. [6]. Paul T. Vianco [7] provided a comprehensive review of different technologies applied to the production of metal-ceramic joints.

Currently, various joining techniques for metal-ceramic joints can be classified into two main categories: mechanical methods (using bolts, screws, or hot-fastening) and chemical-based processes involving interaction between solid or liquid phases. However, mechanical methods have significant drawbacks, especially when operating under cyclic loads. The drilling and threading processes in ceramics require expensive diamond tools to minimize surface defects. Welding (e.g., ultrasound or diffusion bonding) also has its limitations, particularly for producing parts with complex geometries. It is not always possible to guarantee sufficient strength in joints of dissimilar materials, and additional interlayers are often required to reduce residual stress levels. This, in turn, complicates the technological process and increases its cost.

Residual stress occurs during cooling due to the difference in the Young's modulus and the coefficient of thermal expansion (CTE) of the materials being joined. Optimization of a joint's structural design, careful selection of materials, analysis of potential buffer interlayer applications (which can compensate for the difference in material properties), along with thorough testing of brazing parameters, are the primary methods to address the problem of residual stress accumulation [8], [9], [10], [11]. Hence, brazing allows for obtaining joints resistant to cyclic thermal loading, as well as for the production of inexpensive, large-scale parts with complex geometry [11].

One of the classic techniques for obtaining thermally stable metal-ceramic joints is brazing pre-metallized ceramics. This method is a multi-step process. First, the ceramic is coated with a W- or Mo-based paste containing Mn, which is then fired into the surface over several hours of annealing. As a result, a wettable metallic layer is produced on the ceramic, which reacts with the liquefied brazing alloy during the brazing process. However, drawbacks of this technique include various defects occurring during the annealing stage, such as pores, cracks, and uneven sintering of the metallic layer. The high percentage of spoilage caused by these defects, combined with the complexity of the multi-stage production process, makes this an expensive technology. Moreover, it can only be applied to SiO₂-containing ceramics, as the adhesion of the metallic layer is ensured by glass migration through a capillary flow mechanism [12]. Active brazing, a simple and reliable technology, is regarded as a prospective way to solve all the aforementioned problems. It is characterized by the use of brazing alloys containing chemically active elements such as Ti, Zr, Hf, and V. Active brazing alloys used for ceramic brazing are classified into two groups: hard Ti-based alloys [13], [14], [15], [16] and ductile Ag-Cu alloys with active alloying elements [17], [18], [19], [20], [21]. The main difficulty in the production of brazed joints is the accumulation of residual stress, which can cause the failure of the joint.

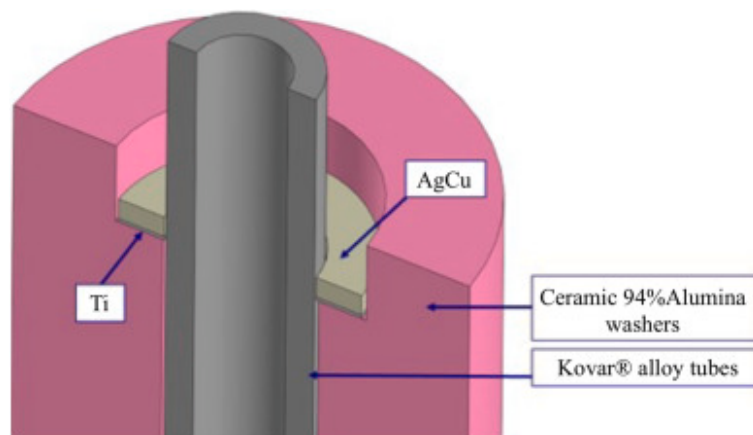
In this work, a eutectic Ag-Cu brazing alloy and titanium foil are suggested for producing a thermally stable joint between Kovar® alloy and 94% alumina ceramic. The thermal stability of the metal-ceramic joint is a key parameter in the design of

modern vacuum units. The most representative criterion of thermal stability is the air-tightness of the joints. However, thermal stability is often studied through microstructural analysis, mechanical testing of the joints, estimation of microstructure evolution (degradation) under thermal loads, and visual assessment of defects [[22], [23], [24], [25], [26], [27], [28], [29]]. The evolution of properties of brazed joints under the influence of thermal loads has been most extensively considered in Ref. [30]. The dependencies determined in that work are of fundamental importance for predicting the properties of metal-ceramic joints produced by pre-metallization.

The aim of the present work is to determine the patterns of the effect of secondary heating on the air-tightness of metal-ceramic Kovar®/94% alumina brazed joints produced by high-temperature brazing using an Ag-based brazing alloy. Furthermore, this study specifically investigates cylindrical joint geometries, a critical yet underexplored design parameter, to establish a foundational understanding of how geometry influences thermal stability.

2. Materials and methods

For the specimens of metal-ceramic joints, Kovar® alloy tubes and 94% alumina ceramic washers were used. The inner diameter of the tubes was 1.2mm and the outer diameter was 1.95mm. The washer diameter was 7mm with a 2.05mm port for the tube, resulting in a clearance of 50µm on each side. The assembly is shown in Fig. 1, and the list of materials is provided in Table 1.



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Fig. 1. Isometry view of the assembly cross-section.

Table 1. Composition of the substrate materials and brazing alloys.

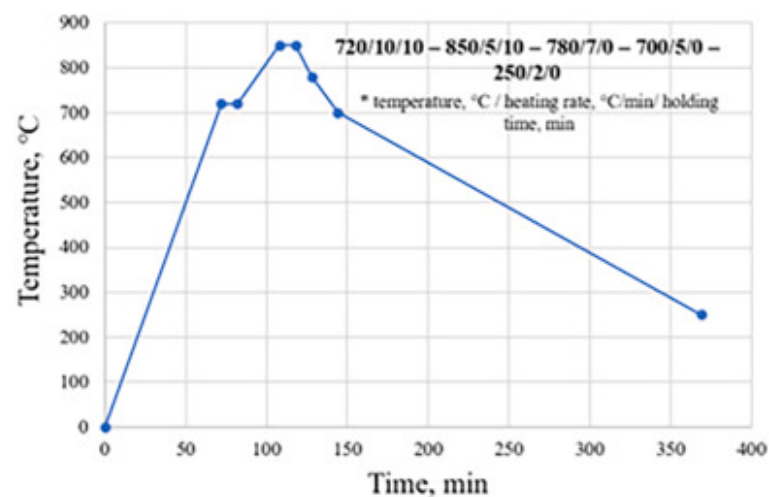
Material	Composition, wt.%
Kovar®	Fe–29Ni–18Co
94%Alumina	Al ₂ O ₃ –2.8SiO ₂ –2.4MnO–0.5Cr ₂ O ₃
AgCu	Ag–28Cu
AgCuTi	Ag–28Cu–1Ti

AgCu and AgCuTi ingots were smelted in an MEPhI-9 argon-arc furnace. The charge materials consisted of copper, titanium, and silver with a purity of 999.9. Rapidly-quenched brazing alloy ribbons, 300 μm thick, were produced using a «Crystall-702» facility via the flat jet casting technique. During casting, the cooling rates reached 10^5 – 10^6 K/s. For brazing, 50 μm thick titanium foil was also used. Titanium inserts were cut from the ribbon and foil by laser-beam cutting. The inserts had an inner diameter of 3.8 mm and an outer diameter of 5 mm.

Substrates for the study of wetting angles were made of 94% alumina ceramic, Kovar® alloy, and titanium. According to GOST 23904-79, it is recommended to use brazing alloy samples of approximately 64 mm³ for the test. To obtain samples of this volume, pieces of about 0.5 g were cut from the ribbon and then spot-welded together. These brazing alloy samples were placed on the substrates and heated in the vacuum chamber of the EP-IRIS installation (an electric furnace for spreading and wetting studies). The pressure did not exceed $\sim 1 \times 10^{-3}$ Pa, the heating rate was 5 °C/min, and the maximum temperature was 900 °C. Cross-sectional images of the brazing alloy droplets on the 94% alumina ceramic, Kovar® alloy, and titanium substrates were recorded with a digital camera during the heating process. The wetting angles were measured using COMPAS-3D software.

The brazing processes for the 94% alumina ceramic and Kovar® alloy were carried out in an SSHVE-1.25/25 vacuum furnace with resistance heating, which maintains a pressure during brazing below 1.3×10^{-3} Pa. The specimens were fastened onto a mount, and the brazing alloy was placed on top of the clearance in a technological opening. All parts were cleaned in alcohol prior to brazing.

The specimens were first heated to a temperature below the solidus point of the brazing alloy at a rate of 10°C/min, with a 15-min holding time to ensure even heating of the mount. Then, the brazing temperature (which was above the liquidus point and varied in the range of 810–850°C) was reached at a heating rate 5–7°C/min the holding time during brazing ranged from 5 to 15 min to allow the clearance to be completely filled with the brazing alloy. Afterwards, the specimens were cooled slowly (at a rate of 7°C/min) to a temperature below the liquidus point but several degrees above the solidus point. A slower cooling rate (5°C/min) was then maintained until the solidus point was reached. Finally, the specimens were cooled to 100°C at a rate of 1–3°C/min. The brazing cycle is presented in Fig. 2.



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Fig. 2. Technological heating and cooling curve for the brazing process.

To study the microstructure and composition of the brazed seams, the specimens were embedded in a polymer resin, ground, and polished using a semi-automated LaboPol-5 Struers grinding and polishing machine. Water-based monocrystalline diamond suspensions and polishing cloths were used for this process. Cross-sectional images of the brazed specimens were obtained using a Carl Zeiss EVO 50 scanning electron microscope. The microstructure was studied via back-scattered electron (BSE) imaging with a four-quadrant BSE detector (4Q-BSD). Compositional analysis was performed with an INCA X-Act energy-dispersive X-ray spectroscopy (EDX) system at a beam accelerating voltage of 5kV.

Thermal cycling testing was carried out in a SPEC MC-811P climatic chamber. A set of specimens was cooled to -65°C and held for 2 h, then heated to $+25^{\circ}\text{C}$ held for 1 h, and finally heated to $+55^{\circ}\text{C}$ and held for 2 h.

Thermal stability is operationally defined as the retention of hermeticity after thermal cycling/reheating, assessed by comparing the post-exposure leak rate to the pre-exposure value and to the applicable acceptance limit from the fine-leak method. Thermal stability was studied on a set of thermally treated specimens. The treatment regime was as follows: heating to $T=450^{\circ}\text{C}/600^{\circ}\text{C}/750^{\circ}\text{C}$ at a rate $v=3-5^{\circ}\text{C}/\text{min}$, holding for 1 h at the target temperature, then cooling to $T=100^{\circ}\text{C}$ at a rate of $v=2^{\circ}\text{C}/\text{min}$. Afterwards, the specimens were furnace-cooled and taken for leak testing.

Additionally, the effect of the heating rate on the air-tightness of the brazed joints was determined. The specimens were heated to 700°C at rates of $7.5^{\circ}\text{C}/\text{min}$, $10^{\circ}\text{C}/\text{min}$, $15^{\circ}\text{C}/\text{min}$, $20^{\circ}\text{C}/\text{min}$, $35^{\circ}\text{C}/\text{min}$, $50^{\circ}\text{C}/\text{min}$, $100^{\circ}\text{C}/\text{min}$, and $150^{\circ}\text{C}/\text{min}$.

Leak testing was performed using a portable Pfeiffer SmartTest helium leak detector with a sensitivity of $10^{-13}\text{Pam}^3/\text{s}$. A specimen was considered air-tight if the registered flow did not exceed the background rate.

Residual stress accumulation in the joint during the brazing process was calculated using ANSYS software. The computational model considered a 94% alumina ceramic/Kovar® alloy joint brazed with an AgCu brazing alloy. The titanium interlayer was not included in the model. The thermo-mechanical properties of the brazing alloy were obtained from references or calculated from experimental data. The mechanical properties of the brazing alloy for elevated temperatures were modeled using JmatPro software. Young's modulus was calculated by the first-principle pseudo-potential method for multicomponent systems under the following assumptions: the material is composed of perfect crystals with coherent boundaries; different phases are distributed evenly, and no texture was present. Elastic behavior was assessed at very low stress levels for several temperature points. Young's modulus and shear modulus were derived from the expression:

$$\frac{E(T)}{E_0} = 1 - 0.2 \cdot \frac{T}{T_{\max}} - 0.25 \cdot \left(\frac{T}{T_{\max}} \right)^2, \quad (1)$$

where E_0 – Young's modulus at room temperature, GPa; $T_{\max}$ – melting point, K.

The thermo-mechanical properties of the substrate materials and the brazing alloy used in the calculation are given in [Table 2](#) for a temperature range of $20-800^{\circ}\text{C}$.

Table 2. Thermo-mechanical properties of the materials.

Temperature, °C	20	100	200	300	400	500	600	700	800
AgCu									
Density, g/cm ³					9.87				
Young's modulus, GPa	83	80	77	73	70	65	60	55	50
Shear modulus, GPa	30	29	28	27	25	24	22	20	18
CTE, °C ⁻¹ , 10 ⁻⁶					18.5				
Kovar®									
Density, g/cm ³	8.3	8.3	8.3	8.2	8.2	8.1	8.1	8.0	8.0
Young's modulus, GPa	210	205	200	192	190	181	172	163	153
Poisson's ratio	0.30	0.31	0.31	0.32	0.32	0.33	0.33	0.34	0.35
CTE, °C ⁻¹ , 10 ⁻⁶	5.4	5.4	4.8	4.6	4.5	6.3	7.8	9.0	9.9
94%Alumina									
Density, g/cm ³					8.2				
Young's modulus, GPa	25	25	24	24	24	24	23	23	22
Poisson's ratio					0.3				
CTE, °C ⁻¹ , 10 ⁻⁶	5.4	5.7	6.0	6.3	6.6	6.9	7.2	7.5	7.8

The mechanical properties used for the bi-linear deformation model are given in.

Table 3. The mechanical properties of the AgCuTi and Kovar® alloys and the 94% alumina ceramic were obtained from references, with their temperature dependencies estimated using equation (1). The tangent modulus was calculated at different

temperatures based on empirical models of its relationship with Young's modulus and yield strength, which depend on the material's hardening behavior, using the Material 2.0 software. The 94% alumina ceramic is a brittle material, meaning its tangent modulus could be neglected.

Table 3. Temperature dependencies of mechanical properties of the materials.

Temperature, °C	20	100	300	500	700	800
Kovar®						
Yield strength, MPa	276	273	263	250	234	225
Tensile strength, MPa	518	511	492	468	439	422
Elongation, %			30			
Tangent Young's modulus, MPa	1325	1304	1255	1195	1122	1079
94%Alumina						
Yield strength, MPa	260	258	252	244	235	230
Tensile strength, MPa	260	258	252	244	235	230
AgCu						
Yield strength, MPa	234	239	227	153	39	1
Tensile strength, MPa	240	253	232	155	40	1
Elongation, %			20			
Tangent Young's modulus, MPa	248	272	239	158	42	0

The properties of titanium were not considered in the calculation, as it dissolves into the brazing alloy during the brazing process.

Norton's approximation (expression 2) was used to estimate the residual stress formed during the cooling stage of the brazing process. The approximation describes the steady-state creep rate:

$$\dot{\epsilon} = A \cdot \sigma^n \cdot e^{-\frac{Q}{RT}}, \quad (2)$$

where $\dot{\epsilon}$ – creep rate; A , n – constant; σ – applied stress; Q – activation energy for creep, R – universal gas constant (8.31 J/mol·K), T – temperature.

The constants A , n , and the Q/R ratio were calculated based on data from Refs. [[31], [32], [33], [34]] for silver and [[35], [36], [37], [38]] for copper.

According to Kornilov [39] and Grigorovych [40], precipitation-hardening alloys with a composition near the solubility limit demonstrate the best performance at elevated temperatures below the eutectic point in binary alloy systems with partial solid solubility. The highest strength at elevated temperatures and the poorest ductility at low temperatures correspond to the eutectic alloy. However, in a temperature range close to the eutectic temperature, such an alloy shows the highest creep rate [40].

In alloy systems with significant component solubility, the "high-temperature strength vs. content" dependence exhibits two distinct peaks at the boundaries of the two-phase field. In contrast, in eutectic systems with narrow α - and β -solid solution regions, the peaks are less pronounced. Between the peaks, the strength values change linearly. In general, the high-temperature strength of a two-phase mixture will increase or decrease depending on whether the high-temperature strength of the second phase is higher or lower than that of the first [39].

Based on the general understanding of the high-temperature strength and creep behavior of eutectic alloys with partial solid solubility, a linear extrapolation was applied to obtain the parameter values used in Norton's approximation. These values are given in Table 4.

Table 4. Parameters used in Norton's approximation.

Temperature, °C	250	400	600
$A (C_1)$	3.00E-24	3.09E-25	2.68E-26

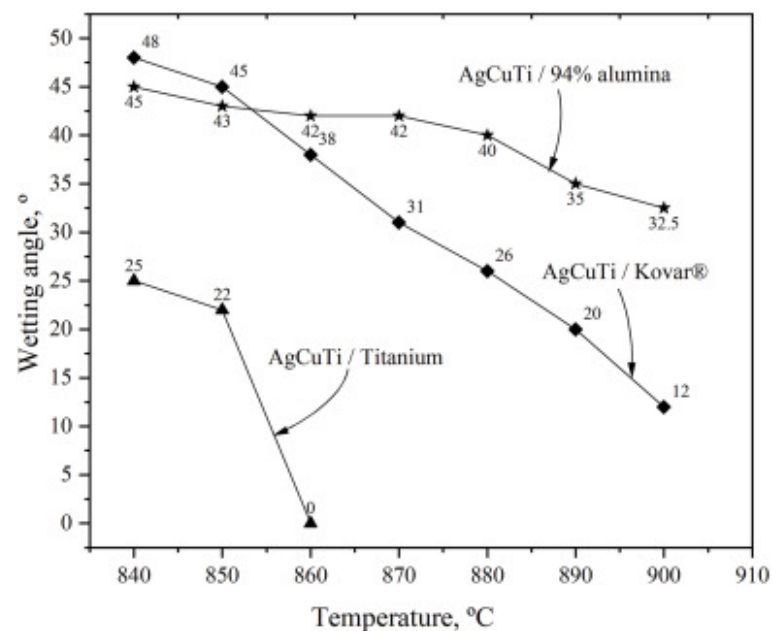
Temperature, °C	250	400	600
$n (C_2)$		4.42	
Q , kJ/mole		188	
Q/R , K		22623	

The computational model considered a cylindrical joint between the 94% alumina ceramic and the Kovar® alloy tube. The solution was calculated for a plane with axial symmetry and a restraint on Y-direction movement applied to the lower flat edge of the ceramic part.

3. Results and discussion

3.1. Wetting characteristics of AgCuTi brazing alloy on different substrate materials

The temperature dependence of the wetting angle of the AgCuTi brazing alloy on different substrates(94% alumina ceramic, Kovar® alloy, and titanium) is shown in [Fig. 3](#). The maximum temperature was 900°C.



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Fig. 3. Wetting angles at different temperatures (840–900°C) in the spreading experiment with AgCuTi brazing alloy and 94%Alumina ceramic, Kovar® alloy, titanium substrates.

It is evident that the AgCuTi alloy starts melting in the 830–840°C temperature range in all three scenarios. However, the minimum wetting angle is reached at different temperatures. Titanium ensures good reactivity with the 94% alumina ceramic, which allows the wetting angle to reach a value of 32.5° at 900°C. When the substrate material is titanium, the brazing alloy becomes enriched with titanium, and complete spreading (0° wetting angle) occurs at 870°C.

However, a brazing alloy with low Ti content exhibits poor wetting characteristics on ceramics. In a series of experiments, a brazing alloy with high Ti content was produced. However, the problem of the alloy/ceramic interaction remained, as titanium significantly raises the liquidus point of the alloy. Therefore, titanium foil was suggested as an interlayer to localize the active chemical element (Ti) at the ceramic interface. A series of experiments with foils of 10, 30, and 50µm thickness was carried out to determine the optimal amount of Ti. It has been established that the production of air-tight metal-ceramic joints requires an

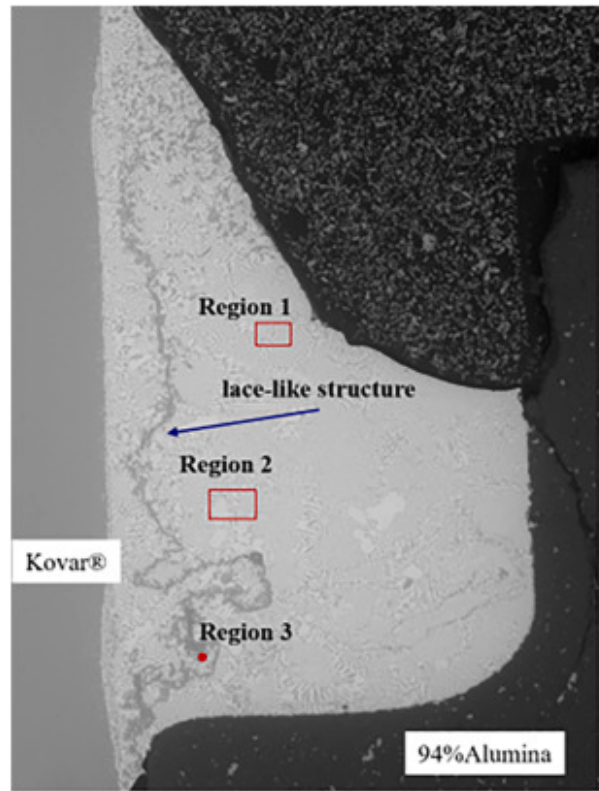
eutectic AgCu brazing alloy 300–900 μm thick combined with a 50 μm thick Ti interlayer. The brazing was carried out at 850 $^{\circ}\text{C}$ with a 10min holding time, according to the brazing cycle shown in [Fig. 2](#). Additionally, a study was conducted to establish the optimal amount of brazing alloy required to obtain thermally stable brazed joints able to withstand rapid temperature changes. The sets of specimens used in the experiments are described in [Table 5](#). The variable in this case is the amount of brazing alloy at the metallic tube interface.

Table 5. List of specimens used in thermal stability studies.

Specimen set №	Amount of the alloy at the metallic tube interface	Number of specimens
1	AgCu (300 μm)	6
2	AgCu (600 μm)	15
3	AgCu (900 μm)	3

3.2. Microstructure investigation

The microstructure of the brazed seam area of a joint brazed with the 300 μm -thick brazing alloy is shown in [Fig. 4](#). EDX results for the seam are given in [Table 6](#).



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Fig. 4. Brazed seam area of the Kovar®/94%Alumina joint, 300μm thick brazing alloy.

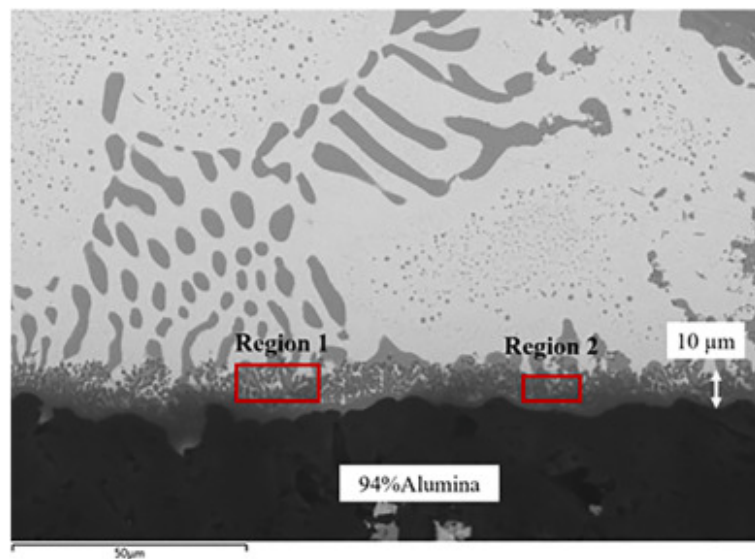
Table 6. EDX results for the brazed seam area of the Kovar®/94%Alumina joint, 300μm thick brazing alloy.

Region	Content, wt. %					
	Ti	Cu	Ag	Fe	Ni	Co
1	0	25.9	69.0	2.8	1.5	0.8
2	0	18.6	74.6	3.8	1.7	1.3

Region	Content, wt. %					
	Ti	Cu	Ag	Fe	Ni	Co
3	24.0	11.2	21.0	17.9	18.8	7.2

According to the EDX results, the structure of the brazed seam consists mainly of the Ag-Cu eutectic (Regions 1, 2). The arrow points to a lace-like structure visible near the Kovar® alloy tube interface. This structure is formed by dissolved titanium (Region 3).

Subsequently, the interaction area between the 94% alumina ceramic and the brazing alloy was studied thoroughly. The microstructure is shown in Fig. 5, and the EDX results are given in Table 7.



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Fig. 5. Interaction area at the 94%Alumina/brazing alloy interface.

Table 7. EDX results for the interaction area at the 94%Alumina/brazing alloy interface.

Region	Content, wt. %						
	O	Al	Ti	Fe	Ni	Cu	Ag
1	15.4	9.7	21.0	0.9	0.7	25.6	26.7
2	21.2	9.1	20.1	0.7	0.6	19.5	28.8

The data in [Table 7](#) allow the area to be divided into two zones. A thin, 10µm-thick intermetallic layer is formed at the ceramic interface. The analysis of Regions 1, 2 suggests the presence of mixed Ti_xCu_xO oxides, [[41](#)], [[42](#)], [[43](#)]].

The mechanism of the interfacial transition layer formation is considered to be as follows After the brazing alloy melts, titanium diffuses to the Al_2O_3 interface and reduces it, forming a layer of titanium oxide (Ti_2O_{1-x}). The reduction reactions are:

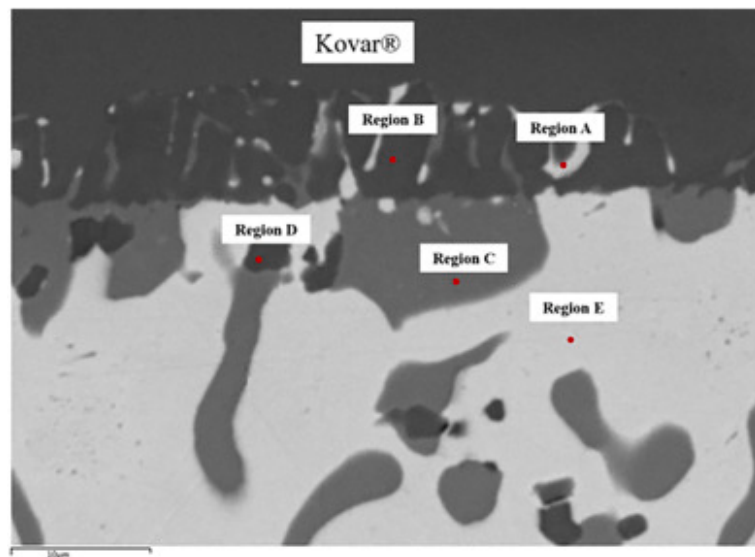


The liberated aluminium dissolves into the silver-rich phase of the braze. In the next stage of the process, the resulting Ti_2O layer acts as a substrate for the formation of a Ti_3Cu_3O layer from the melt according to the reaction:



This sequence of reactions describes the key stages in the formation of a chemically bonded transition zone between the alumina ceramic and the braze alloy.

A similar analysis of the Kovar®/brazing alloy interaction area was performed. The microstructure is shown in [Fig. 6](#), and the EDX results are given in [Table 8](#).



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Fig. 6. Interaction area at the Kovar®/brazing alloy interface.

Table 8. EDX results for the interaction area at the Kovar®/brazing alloy interface.

Region	Content, wt. %							
	O	Al	Ti	Fe	Co	Ni	Cu	Ag
A	5.0	1.6	0.9	21.3	6.4	7.0	7.6	50.2
B	3.1	2.0	1.1	46.4	13.5	13.0	9.0	11.9
C	3.4	2.1	0.6	9.4	2.9	4.8	60.0	16.8
D	4.7	1.7	13.8	31.2	13.7	8.4	9.5	17.0
E	7.8	1.4	0.8	7.9	2.6	3.5	8.2	67.8

The data in Table 8 allow the conclusion that the brazing alloy penetrates along the Kovar® alloy grain boundaries (Regions 1, 3), which leads to the transfer of Kovar® particles into the melt. Region 3 is likely a Cu-based solid solution and Region 5 an Ag-based solid solution. Region 4 is enriched with Ti, indicating the presence of intermetallic compounds. This region might consist of the TiCu and FeTi compounds. Fe- and Ti-based intermetallic compounds peel off from the Kovar® alloy surface and transfer into the melt. This suggestion is supported by the EDX results for Region 4 and is in good agreement with data obtained from Refs. [44,45].

The bonding mechanism between the braze alloy and Kovar® is governed by sequential interfacial reactions initiated by titanium diffusion after brazing alloy melting. The process begins with the formation of a continuous Fe₂Ti layer via the reaction [42]:



This primary layer acts as a diffusion barrier, limiting further growth while providing a substrate for subsequent phase formation. FeTi, Ni₃Ti, and minor CuNiTi phases then nucleate on the Fe₂Ti surface through the following reactions:



Thermodynamic equilibrium conditions favor Ni₃Ti formation over other Ni-Ti compounds at ~800°C, as confirmed by the Fe-Ni-Ti phase diagram.

The FeTi phase forms a secondary layer parallel to Fe₂Ti, further restricting Ti transport. Ultimately, interdiffusion leads to Fe₂Ti layer thickening and FeTi consumption via the reaction:



This sequence of phase formation and transformation establishes the chemical bond at the braze/Kovar interface.

3.3. Thermal stability investigation

Thermal stability testing modes correspond to the operating temperature ranges. The highest possible cooling rate after brazing that allows for the production of airtight brazed joints was established in a rapid furnace-cooling experiment. The obtained value is 12.25°C/min.

The results of the leak test for the different specimen types before and after thermal stability experiments are given in [Table 9](#). All specimens were produced in accordance with the cycle shown in [Fig. 2](#) and utilized a 50µm thick Ti interlayer.

Table 9. Leak test of the brazed joints after heating up to 450–750°C, 1 h holding time.

Set №	Testing modes	Air-tightness, +/-
№1	Heating up to 450°C	+
		+
		+
		+
		+
	Heating up to 600°C	+
		+
		+
		+
		+
	Heating up to 750°C	+
		+
		+
		+
		+

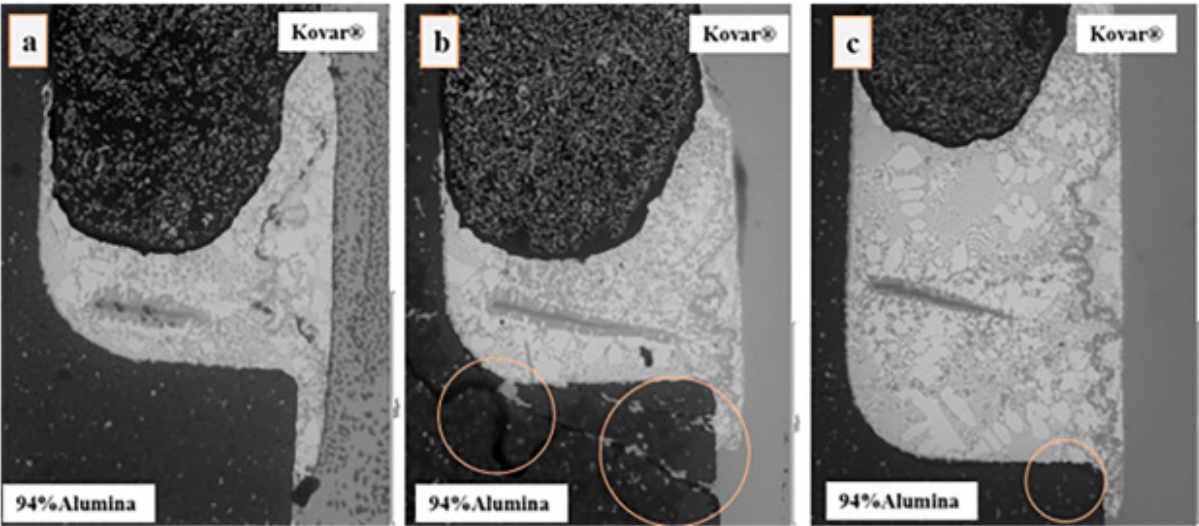
Set №	Testing modes	Air-tightness, +/-
№2	Heating up to 450°C	+
		+
		+
		+
		Major leakage
		+
		+
		Major leakage
		+
		+
		-
		+
№3	Heating up to 600°C	-
		+
		Major leakage
		+
	Heating up to 750°C	+
		+
		Major leakage
		Major leakage

Set №	Testing modes	Air-tightness, +/-
		-
		+

According to the results in [Table 9](#), the first set of specimens demonstrates the best resistance to thermal loading. These specimens remained airtight across all temperature ranges.

Thus, the relationship between the thickness of the Ti interlayer, the Ag-Cu brazing alloy, and the thermal stability of the brazed joint was determined. It was established that the optimal Ti to Ag-Cu ratio is 1:6 by volume, which enables the production of thermally stable and airtight joints.

SEM images of the specimens are shown in [Fig. 7](#), with cracks highlighted by circles. It was determined that these cracks formed in the ceramic during thermal stability testing due to the CTE mismatch between the brazing alloy and the ceramic. Cracks were observed primarily in specimens with a thicker brazing alloy. Thermal stability tests indicated that joints brazed with 600µm and 900µm thick brazing alloys may lose air-tightness, in contrast to specimens brazed with a 300µm thick brazing alloy.



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Fig. 7. SEM images of samples from different sets, with cracks indicated by circles: a – overview of the brazed seam from the first set; b – overview of the brazed seam with cracks from the second set; c – overview of the brazed seam with cracks from the third set.

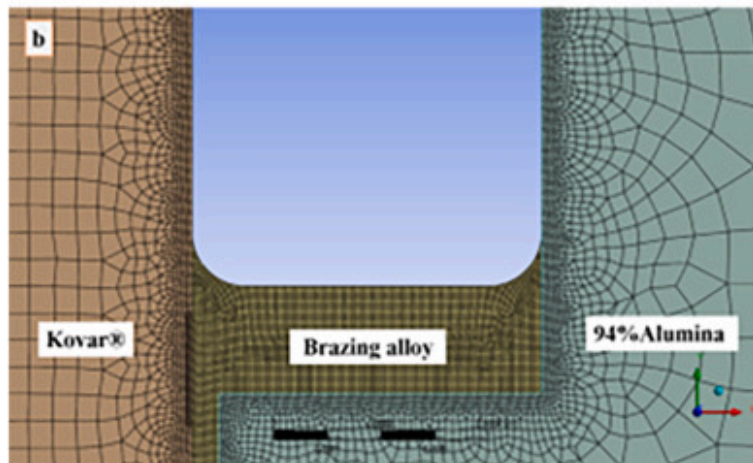
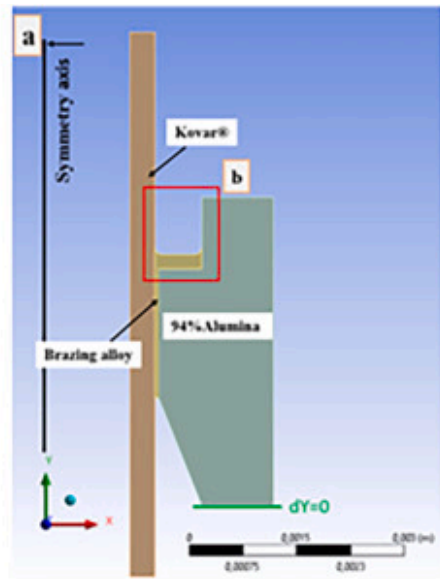
Besides microstructure and processing technology, the properties of brazed joints are also influenced by the geometry and parameters of the fillet region formed by the brazing alloy. The contribution of fillets to the mechanical properties of joints can be up to 70%. Special attention should be given to joints of complex geometry and those with significant CTE differences between the brazing alloy and substrates. An increased fillet size can either improve mechanical properties by increasing the brazed seam area or worsen them due to stress accumulation in one of the joint components. In particular, a change in the fillet angle near the substrate interface can reduce maximum stresses by redistributing them within the brazed seam and substrates, as demonstrated by finite element method (FEM) modeling of homogeneous joints. However, in heterogeneous joints, the redistribution of residual stress may have the opposite effect due to differences in thermomechanical properties, the formation of stress concentrators, and a lever mechanism of stress distribution in parts of complex geometry.

3.4. Numerical modelling of the residual stress in a brazed seam

To establish the relationship between the amount of brazing alloy, stress values, and stress distribution, computational models were created, and numerical modeling of the cooling process of the Tube/Fillet/Ceramic system was performed.

A load was applied to all parts of the assembly through a temperature change from 800°C to 22°C at a cooling rate of 2.5°C/min. The calculations were performed for a radial clearance of 100µm, filled with the AgCu brazing alloy.

The initial conditions set in the Tube/Fillet/Ceramic model are shown in Fig. 8. Fig. 8b shows the mesh in the brazed seam region. The height of the brazing alloy column in the technological opening was set to 200, 400, and 600µm. The fillet radius of the brazing alloy was 200µm at both the 94% alumina ceramic and Kovar® alloy interfaces. The calculated height differs because part of the brazing alloy fills gaps and forms the fillet, thereby reducing the effective thickness. This explains why the values in Section 3.4 (200, 400, and 600µm) are less than the nominal brazing alloy thicknesses (300, 600, and 900µm).

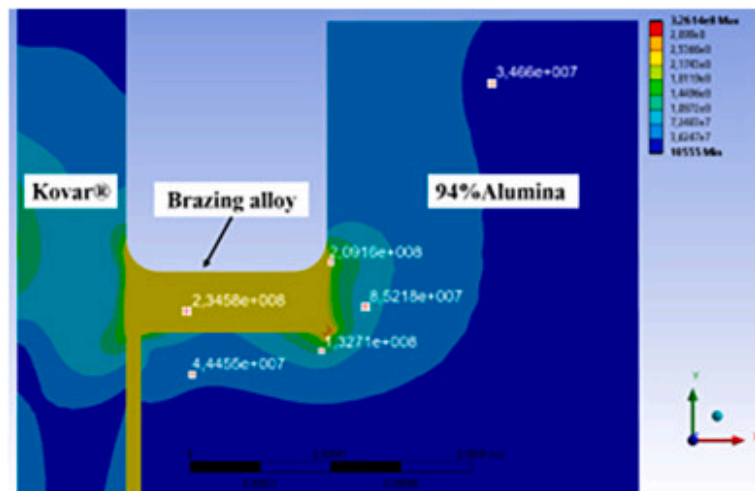
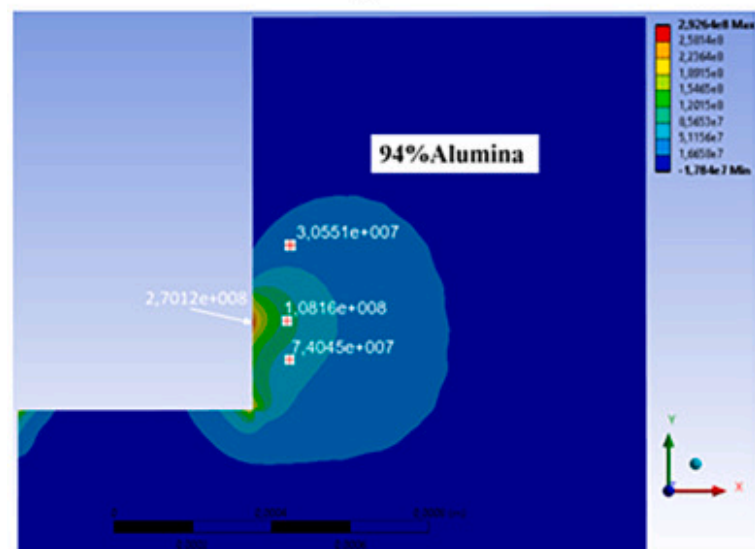


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Fig. 8. Initial conditions and mesh for the Tube/Fillet/Ceramic system: a – general view of the joint, b – mesh in the brazed seam region.

The equivalent von Mises stress distribution was calculated for the Tube/Fillet/Ceramic system. Fig. 9 shows the stress distribution for the 200 μ m high brazing alloy column. The general distribution of the von Mises stress is shown in Fig. 9a. The formation of a low-stress (≤ 50 MPa) field is observed on the ceramic surface facing the gap. Regions of moderate stress (≤ 100 MPa) are formed at the edges of the ceramic part and along the vertical brazing alloy/ceramic interface. At the horizontal interface, the moderate stress distribution is uneven.

**a****b**

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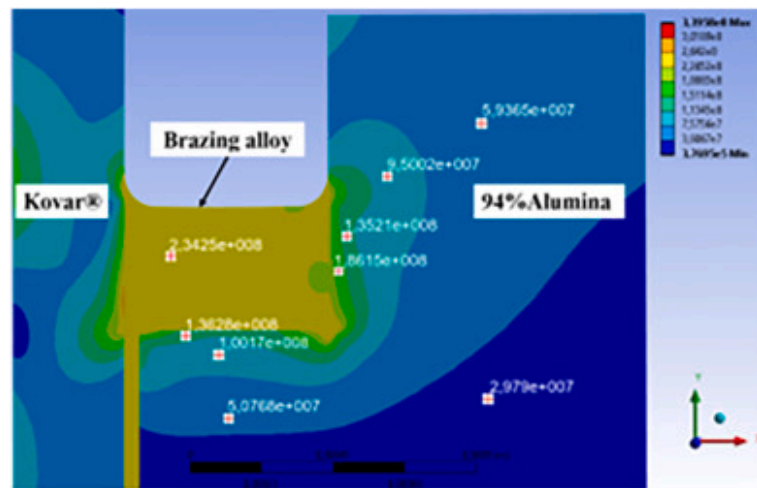
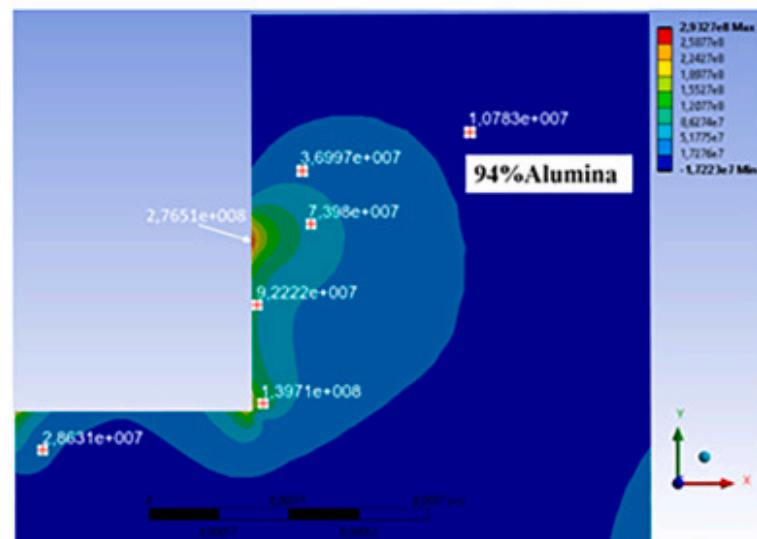
Fig. 9. Stress distribution in the Tube/Fillet/Ceramic system for the 200 μ m high brazing alloy column: a – von Mises equivalent; b – tensile stress in ceramic.

The tensile stress distribution in the ceramic is shown in Fig. 9b. Low tensile stresses (≤ 40 MPa) arise at the edges of the ceramic part. A local stress maximum (above 270MPa) is observed at the fillet/ceramic interface.

Equivalent von Mises stress distribution was calculated for the Tube/Fillet/Ceramic system. Fig. 9 shows the stress distribution for the 200 μ m high brazing alloy column. General distribution of the von Mises equivalents is shown in Fig. 9a. The formation of a low stress (≤ 50 MPa) field is observed on the ceramic surface facing the gap. Moderate stress (≤ 100 MPa) regions are formed at the edges of the ceramic part and along the vertical brazing alloy/ceramic interface. At the horizontal interface, the moderate stress distribution is not even.

Tensile stress distribution in the ceramic is shown in Fig. 9b. Low tensile stresses (≤ 40 MPa) arise at the ceramic part edges. A local maximum (above 270MPa) is observed at the filler/ceramic interface.

Fig. 10 shows the stress distribution for the 400 μ m high brazing alloy column. General distribution of the von Mises equivalents is shown in Fig. 10a. The formation of a low stress (≤ 60 MPa) field is observed at the ceramic surface facing the gap. With the increasing the height of the brazing alloy column from 200 μ m to 400 μ m, low stress intensity goes up by 20% and the corresponding cross-section area increases both in the ceramic and the Kovar® alloy. Moderate stress (≤ 100 MPa) regions are distributed evenly along the ceramic surface facing the brazing alloy. Tensile stress distribution in the ceramic is shown in Fig. 10b. Low tensile stresses (≤ 40 MPa) are distributed evenly along the ceramic surface facing the brazing alloy. A local maximum (above 276MPa) is observed at the filler/ceramic interface.

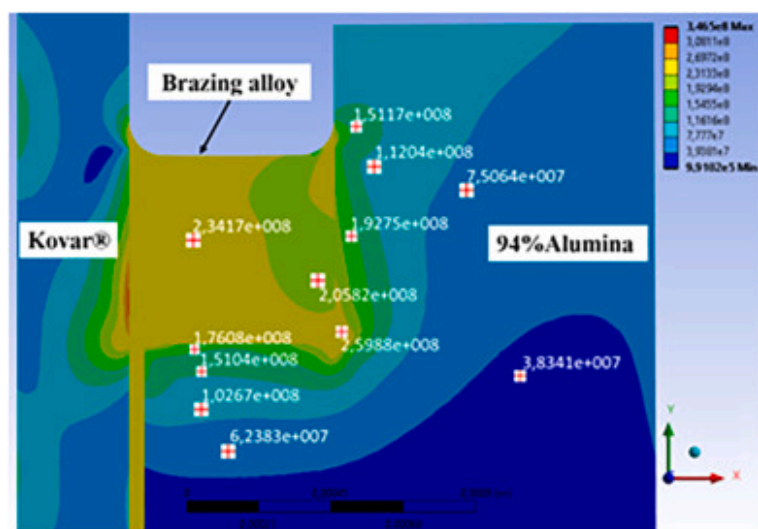
**a****b**

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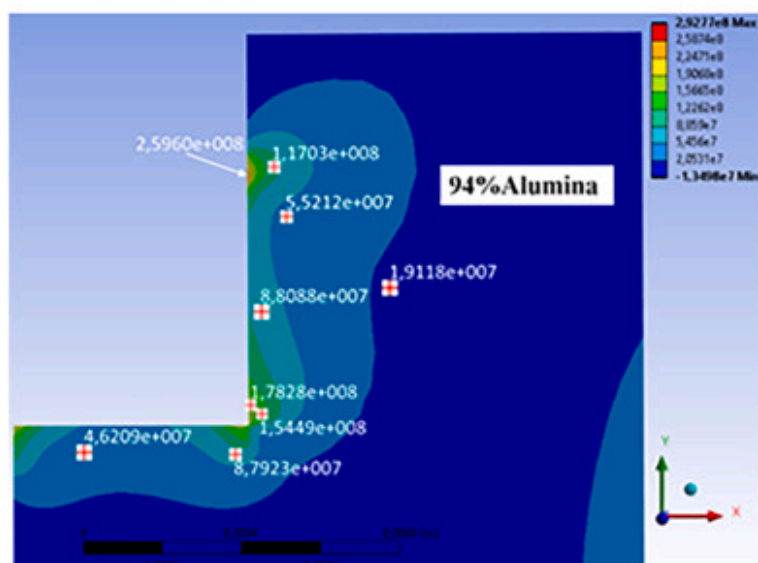
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Fig. 10. Stress distribution in the Tube/Fillet/Ceramic system for the 400 μ m high brazing alloy column: a – von Mises equivalent; b – tensile stress in ceramic.

Fig. 11 shows the stress distribution for the 600µm high brazing alloy column. The general distribution of the von Mises stress is shown in Fig. 11a.



a



b

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Fig. 11. Stress distribution in the Tube/Fillet/Ceramic system for the 600 μm high brazing alloy column: a – von Mises equivalent; b – tensile stress in ceramic.

The low-stress field ($\leq 75\text{MPa}$) shows an increase in stress values (stress intensity rises by 25% as the height of the brazing alloy column increases from 400 μm to 600 μm). The moderate-stress field ($\leq 115\text{MPa}$) expands along the border of the ceramic part facing the brazing alloy, forming a continuous shape that envelops the gap. Additionally, a high-stress field ($\leq 200\text{MPa}$) forms at the ceramic/brazing alloy interface.

The tensile stress distribution in the ceramic is shown in [Fig. 11b](#). Low tensile stresses ($\leq 50\text{MPa}$) are distributed evenly along the ceramic surface facing the brazing alloy. As the height of the brazing alloy column increases from 400 μm to 600 μm , the low-stress intensity increases by 20%, and the corresponding area also expands. A local maximum (above 259MPa) is observed at the fillet/ceramic interface.

An increase in the brazing alloy column height from 200 μm to 400 μm and 600 μm causes an expansion of the elevated stress field, which is concentrated primarily at the brazing alloy/ceramic interface. In addition to the increase in area, the corresponding intensity also rises. As the height of the brazing alloy column decreases, the intensity and area of the tensile stress fields are reduced. This may explain the low thermal stability of joints brazed with the 900 μm thick brazing alloy and the superior thermal stability of joints brazed with the 300 μm thick brazing alloy.

3.5. Thermal cycling testing

Thermal cycling testing was carried out in a SPEC MC–811P climatic chamber on a set of five specimens brazed with a 300 μm -thick brazing alloy and a 50 μm -thick Ti interlayer. The specimens were cooled to -65°C and held for 2h, then heated to $+25^\circ\text{C}$, held for 1h, and finally heated to $+55^\circ\text{C}$ and held for 2h. The results of the leak test for the as-brazed specimens and the specimens after thermal cycling are given in [Table 10](#). The leak detector readings are provided for convenience.

Table 10. Leak test results for the as-brazed specimens and after thermal cycling.

Sample №	Air-tightness as-brazed, Pa·m ³ /s		Air-tightness after climatic chamber test, Pa·m ³ /s	
1	Yes	$2.0 \cdot 10^{-13}$	1	Yes
2	Yes	$2.2 \cdot 10^{-13}$	2	Yes
3	Yes	$3.0 \cdot 10^{-13}$	3	Yes
4	Yes	$4.5 \cdot 10^{-13}$	4	Yes
5	Yes	$5.2 \cdot 10^{-13}$	5	Yes

It is evident from [Table 10](#) that the specimens brazed with the eutectic Ag-Cu brazing alloy and Ti foil successfully passed the test without losing air-tightness.

One of the aims of the study was to determine the critical heating rate for the joint, using a leak test as the criterion for assessing integrity. The heating rate was varied starting from 7.5 °C/min with an increasing step size, as shown in [Table 11](#). For the testing, three metal-ceramic joints brazed with a 300µm-thick Ag-28%Cu brazing alloy and a 50µm thick Ti interlayer were used. There was no holding time at 700 °C, and cooling was initiated immediately at a rate of 2 °C/min.

Table 11. Results of the leak test before and after rapid heating.

Experiment №	Heating rate, °C/min	Air-tightness before testing, Pa·m ³ /s		Air-tightness after testing, Pa·m ³ /s	
1	7.5		$2.0 \cdot 10^{-13}$		$2.3 \cdot 10^{-13}$
			$3.2 \cdot 10^{-13}$		$2.9 \cdot 10^{-13}$
			$5.1 \cdot 10^{-13}$		$4.5 \cdot 10^{-13}$
2	10		$1.0 \cdot 10^{-13}$		$1.0 \cdot 10^{-13}$
			$1.5 \cdot 10^{-13}$		$1.1 \cdot 10^{-13}$
			$1.3 \cdot 10^{-13}$		$1.2 \cdot 10^{-13}$

Experiment №	Heating rate, °C/min	Air-tightness before testing, Pa·m ³ /s	Air-tightness after testing, Pa·m ³ /s
3	15	$3.0 \cdot 10^{-13}$	$1.6 \cdot 10^{-13}$
		$3.8 \cdot 10^{-13}$	$3.5 \cdot 10^{-13}$
		$5.1 \cdot 10^{-13}$	$1.0 \cdot 10^{-13}$
4	20	$3.0 \cdot 10^{-13}$	$1.0 \cdot 10^{-13}$
		$3.8 \cdot 10^{-13}$	$1.1 \cdot 10^{-13}$
		$4.1 \cdot 10^{-13}$	$1.2 \cdot 10^{-13}$
5	35	$1.0 \cdot 10^{-13}$	$1.0 \cdot 10^{-13}$
		$1.0 \cdot 10^{-13}$	$1.0 \cdot 10^{-13}$
		$1.0 \cdot 10^{-13}$	$2.0 \cdot 10^{-13}$
6	50	$1.0 \cdot 10^{-13}$	$1.0 \cdot 10^{-13}$
		$1.0 \cdot 10^{-13}$	$1.0 \cdot 10^{-13}$
		$1.0 \cdot 10^{-13}$	$1.0 \cdot 10^{-13}$
7	100	$1.0 \cdot 10^{-13}$	$1.0 \cdot 10^{-13}$
		$1.0 \cdot 10^{-13}$	$1.0 \cdot 10^{-13}$
		$1.0 \cdot 10^{-13}$	$1.0 \cdot 10^{-13}$
8	150	$1.0 \cdot 10^{-13}$	No, $5.3 \cdot 10^{-8}$
		$1.0 \cdot 10^{-13}$	No, $2.3 \cdot 10^{-6}$
		$1.0 \cdot 10^{-13}$	$1.0 \cdot 10^{-13}$

According to literature sources, it is not advised to heat brazed joints at a rate higher than 5–10°C/min due to the risk of ceramic cracking in the interaction area caused by thermal stress accumulation [30]. However, the data in Table 11 suggest that a rate of up to 100°C/min is acceptable when heating to the joint's operating temperature, and the joint will remain airtight. The fast-cooling experiment after brazing demonstrates that air-tightness can also be maintained for joints that are furnace-cooled. These results can be explained by the deformation ability of the brazing alloy, which allows for stress relaxation.

4. Conclusions

This study investigated the wetting behavior of an AgCuTi brazing alloy on three substrates: 94% alumina ceramic, Kovar®, and titanium. The microstructure of brazed joints, produced using an optimized AgCu alloy with a 50-μm titanium spacer, was analyzed. Additionally, the thermal stability of the joints was assessed through post-treatment leak-tightness testing. Residual stresses induced during brazing were simulated using ANSYS. Key findings include.

- 1) In the temperature range of 800–900°C, AgCuTi brazing alloy exhibits wetting angles of 45.0–32.5° on 94% alumina ceramic and 48.0°–12.0° on Kovar® alloy. Complete spreading (a 0° wetting angle) on the titanium substrate occurs at 870°C.
- 2) The structure of the brazed joints between 94% alumina ceramic and Kovar® alloy consists of a eutectic comprising copper-based and silver-based solid solutions, along with intermetallic compounds (TiCu and FeTi) and a mixed oxide Ti_xCu_xO .
- 3) Hermetic 94% alumina-Kovar® joints were successfully produced by brazing at 850°C for 10 min. Thermal cycling tests, involving cooling to –65°C and heating to +55°C, confirmed that the joints maintained their leak-tightness. Specimens fabricated with a 300μm-thick brazing alloy demonstrated thermal stability upon reheating to temperatures between 450°C and 750°C. Furthermore, these joints retained their seal integrity even when heated at a rapid rate of 100°C/min. The study also established the influence of brazing alloy thickness on resistance to thermal stresses.
- 4) Numerical modeling of the tube/fillet/ceramic system confirmed that the height of the AgCu brazing alloy column significantly affects the stress distribution during cooling. An increase in the brazing alloy column

height from 200µm to 600µm raises the stress intensity by up to 25% and expands the high-stress zones, particularly at the ceramic/brazing alloy interface.

CRedit authorship contribution statement

S.M. Terekhova: Writing – original draft, Visualization, Investigation. **A.A. Ivannikov:** Supervision, Methodology, Conceptualization. **A.V. Abramov:** Writing – review & editing, Software. **V.O. Kirillova:** Writing – review & editing. **O.N. Sevryukov:** Project administration.

Statements and declarations

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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